Jihyeong Jeon

Ph.D. in Artificial Intelligence
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Research Interests

- **Financial AI & Quantitative Research:** Portfolio Optimization, Risk-Aware Asset Allocation, Financial Time-Series Modeling, Robust Prediction under Distribution Shift, Quantitative Trading
- **Natural Language Processing & Information Retrieval:** Large Language Models, Ambiguity-Aware QA, Retrieval-Augmented Generation (RAG), Personalized Retrieval, Generative Retrieval
- **Machine Learning & Deep Learning:** Reinforcement Learning, Diffusion Models, Representation Learning, Multi-Agent Reinforcement Learning, Transformers

Education

Seoul National University, Seoul, Republic of Korea — Aug. 2026
Ph.D. in Artificial Intelligence
Graduate School of Artificial Intelligence
Advisor: Prof. U Kang
Dissertation: *Balancing Return and Risk in Portfolio Optimization*

Purdue University, West Lafayette, IN, USA — Dec. 2019
B.S. in Mathematics & Computer Science
Department of Mathematics & Department of Computer Science

Professional Experience

DeepTrade Technologies — Jun. 2021 – Nov. 2022
AI Research Engineer

- Evaluated systematic investment models and alpha signals using historical market data, assessing return performance, stability, and downside risk.
- Built backtesting and performance-analysis workflows incorporating precise transaction costs and slippage; analyzed cumulative return, maximum drawdown, volatility, Calmar ratio, Sharpe ratio, and related risk-adjusted metrics.
- Compared alpha signals across market regimes and parameter settings to identify performance drivers, assess robustness, and inform strategy refinement and practical deployability.

Publications

Learning Diagnostic Clarifying Questions for Conversational Question Answering via Data-Efficient Dynamic Distillation — **EMNLP 2026**
Jihyeong Jeon, Cheol Ryu, and U Kang
Conference on Empirical Methods in Natural Language Processing, Budapest, Hungary

Diffusion Models for Risk-Aware Portfolio Optimization — **WSDM 2026**
Jihyeong Jeon, Jeongyoung Lee, and U Kang
ACM International Conference on Web Search and Data Mining, Boise, Idaho, USA

Entity-Aware Generative Retrieval for Personalized Contexts — **CIKM 2025**
Jihyeong Jeon, Jiwon Lee, Cheol Ryu, and U Kang
ACM International Conference on Information and Knowledge Management, Seoul, Republic of Korea

Mitigating Distribution Shift in Stock Price Data via Return-Volatility Normalization for Accurate Prediction **CIKM 2025**
Hyunwoo Lee, **Jihyeong Jeon**, Jaemin Hong, and U Kang
ACM International Conference on Information and Knowledge Management, Seoul, Republic of Korea

FreQuant: A Reinforcement-Learning based Adaptive Portfolio Optimization with Multi-frequency Decomposition **KDD 2024**
Jihyeong Jeon, Jiwon Park, Chanhee Park, and U Kang
ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, Barcelona, Spain

Accurate Stock Movement Prediction with Self-supervised Learning from Sparse Noisy Tweets **BigData 2022**
Yejun Soun, Jaemin Yoo, Minyong Cho, **Jihyeong Jeon**, and U Kang
IEEE International Conference on Big Data, Osaka, Japan

Research & Industry Projects

Ambiguity-Aware Query Understanding for On-Device Language Models May 2024 – May 2026
Samsung Electronics MX — Research Project Lead
Led two industry-academic projects on ambiguous-query understanding: disambiguation-aware retrieval for RAG and clarifying-question generation for conversational QA. Defined research problems and team directions while contributing methods, leading to a CIKM 2025 and EMNLP 2026 publications.

Multi-Agent RL for AI Aircraft Simulation Mar. 2024 – Feb. 2025
Korea Aerospace Industries / ADD / Republic of Korea Air Force — Research Project Lead
Led MARL development for simulated AI aircraft agents, guiding reward design, hierarchical and self-play training, policy deployment, and technical reporting under scenario-specific constraints.

Project–Freelancer Matching Recommendation Apr. 2023 – Aug. 2024
eLancer — Research Project Lead
Led development of a project–freelancer recommender combining text and profile representations with LightGCN-based collaborative filtering; directed model design, feature engineering, and evaluation.

Personalized Plan and Schedule Recommendation Dec. 2022 – Apr. 2023
Samsung C-Lab — Research Advisor
Advised the design of a personalized plan and schedule recommendation system integrating content-based filtering, user-interaction signals, and sequential behavior modeling with RNNs.

AI-Based Patent Management Oct. 2020 – Sep. 2021
Hanwha Ocean Co., Ltd. — Researcher
Developed NLP and text-mining methods for large-scale patent similarity search, automatic technical classification, and temporal analysis of patent-claim topics.

Teaching Experience

· Quantitative Trading Algorithms @ SNU KDT BigData Fintech — Aug. 2026
· Deep Learning @ Samsung C&T — Apr. 2026
· Deep Learning & Machine Learning @ Samsung SDS — Jan. 2026
· Machine Learning @ Hyundai Motor — Jul. 2024, Oct. 2024, Jun. 2025
· Mathematics for Machine Learning @ HD Hyundai — Jun. 2024
· Deep Learning @ LG DS — Feb. 2021, Feb. 2024
· Introduction to IoT, AI & Big Data (Practicum) @ SNU — Mar. 2023
· Reinforcement Learning @ SNU — Mar. 2021
· Deep Learning @ Samsung DS² — Oct. 2020

Professional Services

Reviewer

· KDD, NeurIPS, ICLR, ICML — 2024 – Present

Invited Talks

· KCC 2026 — Top Conference Session — Jun. 2026
· KSC 2025 — Top Conference Session — Dec. 2025
· KSC 2024 — Top Conference Session — Dec. 2024

Patents

· Method for Generating Responses to Queries, Electronic Device Performing the Method and Storage Medium — May 2026
KR Patent Application No. 10-2026-0131731

· Electronic Device and Search Method Using Personalized Entity Aware Retrieval System — Jan. 2026
KR Patent Application No. 10-2026-0002605

· Method and Apparatus for Predicting Financial Data Based on Return-Volatility Normalization to Mitigate Distribution Shift of Financial Data — Aug. 2025
KR Patent Application No. 10-2025-0122710